

Characterization of Macropore Flow Mechanisms in Soil by Means of a Split Macropore Column

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ABSTRACT

Naturally occurring macropores such as root channels and earthworm holes have been shown to affect water flow and solute transport processes in soils. To further our understanding of the specific processes that are involved in macropore flow, we have developed a Split Macropore Column (SMC) which consists of a semi-cylindrical soil column with a semi-cylindrical macropore at the main axial of its planer surface. The SMC allows one to control a number of important physical parameters such as the size, density, and continuity of the macropores for miscible displacement studies, as well as an opportunity to observe visually the macropore's impact on flow processes at the critical matrix-macropore interface. After rigorous testing, we found that this system is a useful instrument to study macropore flow mechanisms. Using these SMCs, we performed a series of miscible displacement experiments to study the effects of soil type, flux rate, macropore size, and macropore to matrix size ratio. Our initial results show three zones of flow through the column: (i) film flow at the matrix-macropore interface, (ii) in the matrix directly adjacent to the macropore, and (iii) in the remaining matrix. The initial data suggest that macropore flow is influenced more by the matrix-macropore size ratio than by the macropore size.

MACROPORES are a natural and ubiquitous component of the soil environment and are known to contribute to enhanced flow of water, solutes, microorganisms, and particulate matter. Several thorough reviews documenting the occurrence and effects of macropore flow have previously been published (Thomas and Phillips, 1979; White, 1985; Germann, 1990; Shipitalo and Edwards, 1996). However, because of the complex nature of soil macroporosity, limited mechanistic information regarding the processes which govern macropore flow is available (Beven and Germann, 1982; Glass et al., 1995; Li and Ghodrati, 1997; Tokunaga and Wan, 1997).

Many of the mechanistic studies of macropore flow in soil have been performed in quasi-natural systems such as undisturbed soil columns or columns seeded with earthworms or root systems. These studies have been successful in demonstrating that hydrologically active macropores can be formed by earthworm activity, the degradation of roots, or simply by a continuous network of interpedal zones (Bouma and Anderson, 1977; Booltink and Bouma, 1991; Li and Ghodrati, 1994, 1995). Other studies have focused on specific aspects of macropore flow dynamics such as the effects of tillage, pore continuity, soil surface coverage, precipitation intensity, and antecedent water content (i.e., soil wetting

history) (Andreini and Steenhuis, 1990; Shipitalo et al., 1990; Edwards et al., 1993; Granovsky et al., 1994; Shipitalo and Edwards, 1996).

Another group of studies has used artificial systems to evaluate macropore flow processes. These systems include uniformly packed soil matrices containing constructed macropores (Czapar et al., 1992; Ela et al., 1992; Heathman et al., 1995; Li and Ghodrati, 1997), or completely artificial matrix-macropore systems such as glass matrix-macropore systems (Wan et al., 1996), and porous plate-capillary tube systems (Phillips et al., 1989). Artificial systems have been invaluable in testing theories regarding macropore flow in soils. The uniform matrix-constructed macropore systems (UMCM) have shown that bi- and poly-modal breakthrough curves (BTC) can be produced by transport through macropores during miscible displacement experiments. This phenomenon has been seen in several field experiments (Ehlers, 1975; Currie et al., 1979; Jury et al., 1986; Jabro et al., 1991; Flury et al., 1994; Tsuboyama et al., 1994) and undisturbed soil column experiments (Elrick and French, 1966; Cassel et al., 1974; Bouma and Dekker, 1978; Mallants et al., 1994; Ward et al., 1994; Heathman et al., 1995; Li and Ghodrati, 1997).

Several authors have suggested that, in addition to surface connected macropores, subsurface macropores play an important role in the preferential flow of water and solutes (Quisenberry and Phillips, 1976; Li and Ghodrati, 1997). Using their glass micromodel method, Wan et al. (1996) also showed the ability of subsurface macropores to transmit water and solutes. None of these studies were able to quantify mass transport through macropores. However, Li and Ghodrati (1997) noted significant increases in saturated hydraulic conductivity (K_s) when macropores were present and that macropore flow initiated below the assumed K_s . In contrast, Czapar et al. (1992) did not observe transport through subsurface macropores.

The artificial systems of Phillips et al. (1989) and Wan et al. (1996) have provided several insights to the mechanisms and dynamics of macropore flow. Using their pressure plate/capillary tube system, Dunn and Phillips (1991) showed that the fraction of hydrologically active macropores (i.e., effective macroporosity) is much less than the actual macroporosity of the soil. By the same system, it was also shown that water can move through macropores under a negative matric pressure (Phillips et al., 1989). This work was further strengthened by Wan et al. (1996) who also demonstrated that in their micromodel system, macropore flow occurred under slightly negative matric pressures while

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the matrix was nearly saturated. These authors also demonstrated the dynamics and continuity of the macropore water film during macropore flow. Their findings support the theory postulated by Quisenberry and Phillips (1976) that a continuous film of water is necessary to initiate macropore flow.

The majority of studies performed in UCM type systems have noted several limitations. These include macropore configurations which are subject to change, become clogged, or even collapse during the experiment (Czapar et al., 1992; Ela et al., 1992; Heathman et al., 1995; Li and Ghodrati, 1997). This limits the ability to quantify the macropore size, shape, density, etc. Further, because of their “black box” nature, these systems do not permit examination of the interfacial dynamics (such as mass exchange rates) between the matrix and macropore. To investigate the nature of this mass exchange between a macropore and its surrounding matrix, a system must be used which (i) allows the use of a variety of different soil types or porous media, (ii) permits access to the macropore-matrix interface for direct examination of all transport processes which are occurring at that interface, and (iii) assures the integrity of the macropore structure at all water regimes.

To meet these challenges, we have developed a simple system with which to study matrix-macropore interfacial dynamics. The split macropore column (SMC) is a semi-cylindrical column containing a macropore of fixed diameter. The SMC allows one to control a number of important physical parameters such as the macropore size, density, and continuity for displacement studies, and provides an opportunity to visually observe the macropore's impact on flow processes at the critical matrix-macropore interface. Surface runoff can also be included or easily removed—with certainty—from the macropore component. Furthermore, the integrity of the macropore is maintained by a mesh screen and the homogeneity of the matrix flow can be monitored. The boundary conditions of this system are such that water enters the upper boundary of the column at a constant rate; the system is at steady state and no surface runoff into the macropore occurs.

Objectives

The objectives of this study were to develop a novel system to examine the interactions between a single macropore and the surrounding soil matrix. Our goals for this system are to characterize the effects of soil type, flux rate, macropore size, and macropore to matrix size ratio. This system has been rigorously tested and preliminary results regarding conditions where macropore flow occurs are presented.

MATERIALS AND METHODS

The Split Macropore Column

Conceptually, a simple soil macropore system can consist of a cylindrical column which contains a single cylindrical macropore at its main axis. Slicing such a column longitudinally through its main axis (i.e., with an imaginary plane) would yield two semi-cylindrical soil columns containing semi-

cylindrical macropores at their centers. Because of the complete symmetry of the sliced system, transport processes in either of the two halves or the whole column are theoretically identical. It also follows that during a miscible displacement study, the impact of a semi-macropore to a half-column matrix would be nearly identical to the impact of a full macropore to a whole-column matrix. Accordingly, in place of a whole column, one could theoretically perform a typical miscible displacement study in a semi-cylindrical column without altering any of the transport parameters. One of the advantages of such a system is that the sliced planar surface “opens” the soil system and allows one to make direct, visual observations (and/or measurements) at the critical interface between the macropore and the soil matrix.

To use the SMC for mechanistic studies of macropore flow, the system must satisfy several constraints. Specifically, to be able to build upon the existing database of macropore flow experiments, this semi-cylindrical system must have the same flow characteristics as a typical cylindrical column. As a result, the semi-cylindrical geometry and macropore screen mesh should have no effect on flow through the system or interfere with packing the column homogeneously.

The Split Macropore Column shown in Fig. 1 is based on these principles. It consists of a semi-cylindrical plexiglas tube with a flat plexiglas plate covering its vertical planar surface. The planar surface has a continuous longitudinal slit along its center representing the open planar surface of the semi-cylindrical macropore. To produce the semi-cylindrical boundary of the macropore and to prevent its possible collapse

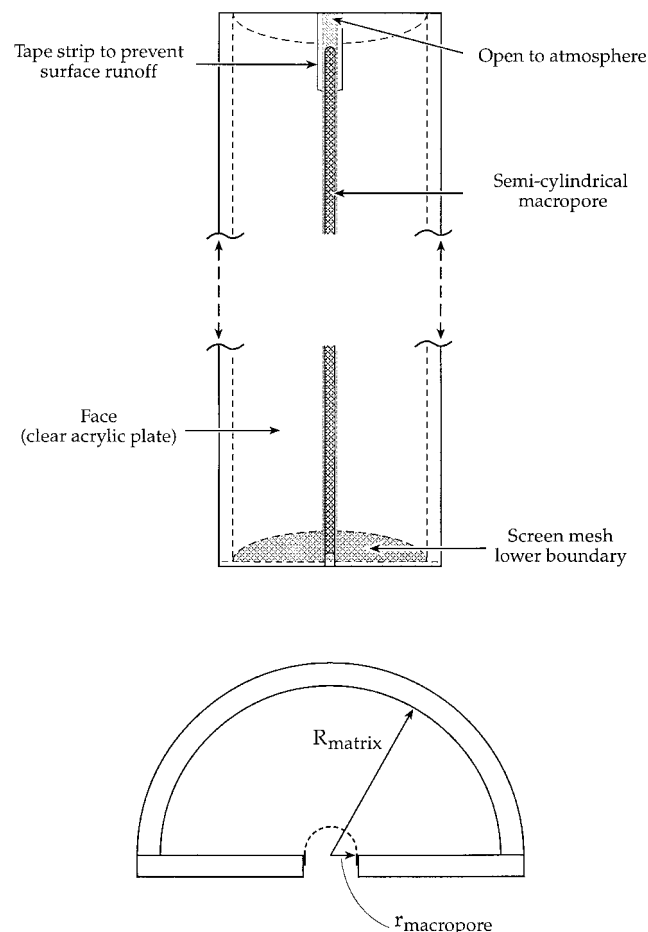


Fig. 1. Diagram of the Split Macropore Column (SMC) showing the face and top view (not to scale).

during miscible displacement studies, a 30-mesh stainless steel screen shaped into a semi-cylinder, is attached to the inside of the slit. The relatively large openings of the 30-mesh screen allow easy passage of water and air while preventing collapse of the macropore. The bottom of the Split Macropore Column is formed with the same stainless steel screen mesh producing a matric potential of zero (atmospheric) at the boundary.

The dependence of macropore flow on macropore size and on matrix to macropore ratio can be examined by constructing a number of columns with different macropore radii (r) and column or matrix radii (R). The macropore runs throughout the length of the column but can easily be sealed from inside with transparent tape to vary the effective length and continuity of the macropore and to avoid surface runoff. For the results presented here, four different columns were used with two matrix radii (R) of 4.5 and 2.5 cm and two macropore radii (r) of 0.32 and 0.16 cm. All four columns were 45 cm long. Terminology is listed in Table 1. The column cross-sectional area was determined by measuring the column volume and dividing by the column length. First, the macropore and the bottom of the column were sealed. Then a known volume of water was added to a known height within the column, so that the cross-sectional area of the column could be calculated (Table 1).

Soils

A commercially available silica sand (90.5% SiO_2) or a Botella clay loam (Mixed, thermic, Typic Xeropsamment) was used for all experiments presented. For the silica sand experiments, the bulk density (ρ_b) was $1.54 \pm 0.01 \text{ g/cm}^3$ and the volumetric water content (θ_v) was $0.36 \pm 0.02 \text{ m}^3/\text{m}^3$. The Botella clay loam had a bulk density (ρ_b) of 1.25 g/cm^3 and a volumetric water content (θ_v) of $0.47 \text{ m}^3/\text{m}^3$. Water content was measured gravimetrically before and after each experiment. Percentage saturation for the silica sand and Botella clay loam was calculated to be 86 and 89%, respectively, assuming a particle density of 2.65 g/cm^3 . Silica sand was chosen for the majority of experiments, specifically the system evaluation, due to its high hydraulic conductivity. In addition, since the silica sand is a light colored soil, movement of the red dye tracer was much easier to observe than in a darker soil, such as the Botella clay loam.

System Evaluation

During the development of this system, several experiments were conducted to test the validity of using this system for studying macropore flow processes. Some of the possible problems we considered in constructing and testing the column were (i) flow instabilities as a result of non uniform column packing; (ii) heterogeneous flow due to sharp edges at the screen mesh/column junction and where the planar and cylindrical surfaces meet (Fig. 1); and (iii) flow along the screen mesh surface rather than as a result of the macropore itself. Four tests were performed on the split macropore column to determine the effects of packing, macropore edge, and

macropore screen on preferential flow. Red dye (FD&C red #40) was used as a tracer since its transport is visually observable. In silica sand, the red dye exhibited little retardation and was assumed to be a conservative tracer for the purposes of visual observations. During these experiments, the large/large SMC ($R = 4.5 \text{ cm}$, $r = 3.2 \text{ mm}$) was used and the influx rate was set to the saturated hydraulic conductivity of the matrix-macroporous system (J_p , see below). The entire macropore screen was sealed from the inside with clear tape. First different packing procedures were tested to determine which procedure produced uniform flow through the column. Development of an appropriate packing method allowed us to test the effects of the macropore edge on the production of preferential flow while the macropore was sealed.

Two experiments were then performed to determine the effect of the macropore screen on macropore flow. The first involved a miscible displacement in a silica sand packed SMC with a detachable screen where the screen was removed after the column had reached a steady state. In the second experiment, the Botella clay loam was used to seal the macropore from outside. The Botella clay loam was chosen because of its relatively fine texture (25% clay) to seal the macropore but to still allow interaction between the sand matrix and macropore screen. First, the clay loam was dry packed into the macropore from the outside of the column, then carefully wetted with a water spray. Finally, transparent tape was applied over the clay loam and surrounding plexiglas to minimize readjustment of the soil during the experiment.

Column Packing

Columns were packed uniformly to a height of 42 cm. Prior to packing, the top 4 cm (from 40–44 cm from the bottom of the column) of macropore screen was sealed from inside (Fig. 1 and 2). As a result, the top 2 cm of the soil surface did not have an open macropore. This was done to ensure that the surface added water will flow only through the matrix (in this case for 2 cm) before encountering any macropore.

A great deal of preliminary work went into ensuring that the SMC was uniformly packed. It should be stressed that nonuniform packing of any soil column can easily lead to erroneous results, and the impact can be quite serious in studies that are investigating preferential flow (Fig. 2a). From our initial packing methods, initial dye transport experiments resulted in non uniform flow at the column edges and the matrix-macropore interface (Fig. 2a). We noted several possible causes for this behavior and ultimately we decided upon

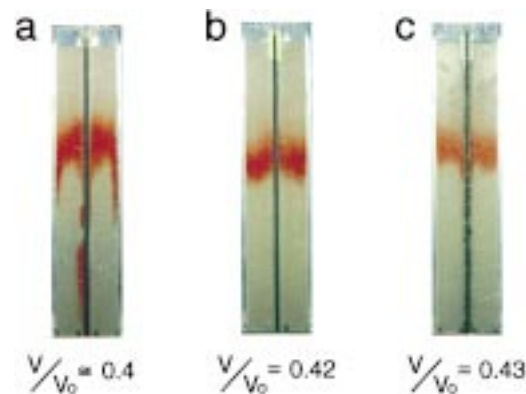


Fig. 2. Influence of the SMC on producing preferential flow patterns. Dye transport patterns at J_p where a) the column is poorly packed (depth is approximate), b) the macropore is sealed from inside, and c) the macropore screen is sealed using clay loam soil. V/V_0 represents the number of pore volumes.

Table 1. Split macropore column sizes.

SMC	Matrix size (R)	Macropore size (r)	Surface area	$J_p^\dagger$	
	cm	mm	cm^2	cm/h	10^{-5} m/s
Large/large	4.5	3.2	30.3	24.1 ± 0.4	6.7 ± 0.1
Large/small	4.5	1.6	31.8	27 ± 3	7.5 ± 0.8
Small/large	2.5	3.2	10.5	34 ± 5	9.4 ± 1
Small/small	2.5	1.6	10.6	33 ± 5	9.2 ± 1

$^\dagger J_p$ values are for the silica sand and include the standard deviation.

a procedure which we found to produce excellent soil homogeneity and be easily reproducible. Prior to packing we taped the macropore from the outside and then carefully removed the tape just after the column was wet. This was done to prevent soil from exiting the macropore during the packing process. The packing procedure involved sprinkling the soil (either through a narrow neck funnel or some other perforated container) into the column from above. We found that it was important to sprinkle the soil slowly (e.g., 4 cm/min) and uniformly without tapping or shaking the column. To ensure that the particles randomly oriented themselves, the funnel was shaken and moved over the column creating a “dust storm” within the column. The funnel size was chosen so that no visible separation of particle sizes could be seen as the media exited the funnel. As a result of this packing method, the porous media was packed to a bulk density less than what we targeted. To achieve our desired bulk density we uniformly tapped (i.e., dropped) the column onto a flat rubber sheet from a height of approximately 5 cm above the sheet. The tapping was repeated about 10 times. We tested this procedure by sealing the macropore from the inside of the column and found that the flow patterns were very homogenous (Fig. 2b).

Tracer Application and Analysis

In addition to the red dye, both nitrate and bromide were used as conservative water tracers. A multi-channel peristaltic pump (ISMATEC, Wallington, England)¹ was used to establish a steady flux in the soil columns. The irrigation channels were distributed to approximate a uniform dripping of solution onto the surface of the columns. Four channels were used for the small matrix columns and seven channels were used for the large matrix columns. The columns were leached with a solution of 0.006 M CaCl₂ in distilled water (Klute and Dirksen, 1986). The tracer solution was applied as a 1-cm pulse (i.e., tracer pulse volume = column surface area × 1 cm) such that a type III boundary condition was used (van Genuchten and Alves, 1982).

During miscible displacement experiments, the column effluent was continuously sampled and collected with a funnel with a side stem. By tilting the funnel slightly, the side stem acted as a reservoir for the column effluent. Small diameter tygon tubing (0.08 cm) was connected to the stem and a variable speed peristaltic pump continuously removed a fraction of the effluent to be analyzed. For columns containing silica sand, the effluent was then passed through a UV absorbance detector (ISCO, Inc., Lincoln, NE) where the solution was optically analyzed for the tracer. Optical measurements of the effluent were converted to nitrate concentration using a measured linear calibration curve ($r^2 = 1.00$) (Chendorain and Ghodrati, 1999). For the Botella clay loam experiments, bromide was used as a conservative tracer. The effluent was sampled and collected using a fraction collector (ISCO, Inc., Lincoln, NE) and analyzed with a bromide selective probe (Fisher Scientific Company, Pittsburgh, PA). Bromide was used as a tracer in the Botella clay loam because the low background concentration of bromide in the clay loam enhanced the sensitivity for measuring macropore effluent diluted into the total effluent. The initial concentrations of nitrate and bromide were both 10 g/L.

¹ Mention of trade names are solely to provide specific information and does not constitute an endorsement over similar products not mentioned.

Calculation of Macropore Hydraulic Conductivity (J_p) of the Soil–Macropore System

After steady state was reached (when influx equaled outflux), the initial flux rate was increased to the highest flux rate possible before water visually ponded on the surface (J_p). Similar to Li and Ghodrati (1997), we have defined this flux rate where ponding began as the approximate saturated hydraulic conductivity of the soil–macropore system.

Macropore Flow Experiments

Experiments were conducted with the SMCs and silica sand to determine under what conditions preferential flow occurs in a single macropore system. Once a steady state had been reached in each column, the J_p was determined (Table 1). Then a series of miscible displacement studies were performed with red dye in the large matrix, small macropore column and the large matrix, large macropore column at fractions of J_p : approximately 33, 66, and 100% J_p . This series of experiments was repeated with nitrate as a conservative water tracer in newly packed columns. A series of experiments was also performed in each of the four columns at J_p and at 1-cm ponded water with nitrate as the tracer. A number of miscible displacement experiments were performed with Botella clay loam soil to ensure that the occurrence and/or nature of macropore induced flow observed were not unique to silica sand.

RESULTS AND DISCUSSION

System Evaluation

Aside from its specific function as a system for studying macropore flow processes, the SMC, because of its semi-cylindrical shape, proved to be an excellent device for testing methods to uniformly pack soil columns. The transparent planar surface allowed us to view any non-uniformity occurring between the column edge and its interior. This “visibility” and our ability to easily seal the macropore, allowed us to determine the most appropriate packing procedure for our experiments (as described above). Figure 2b further illustrates that when the macropore is sealed and the matrix is uniformly packed, no preferential flow results from interactions with the macropore edge as it meets the column planar surface.

The effects of the screen mesh were tested in two ways. Using the removable screen SMC, we observed that macropore flow patterns were similar to those seen in columns where the screen was left intact (data not shown). However, removal of the screen always resulted in the eventual collapse of the macropore. When the clay loam soil was packed into the macropore, dye movement through the column was uniform (Fig. 2c). Hence, the screen mesh itself did not produce any significant effect on the initiation and propagation of flow through the macropore. After performing these evaluations and approximately 75 recorded dye transport experiments to test the system, we concluded that the SMC system does not artificially produce preferential flow or interfere with flow produced by the macropore. Further, we concluded that the SMC is an adequate system to study the interactions between a macropore and a homogenous matrix.

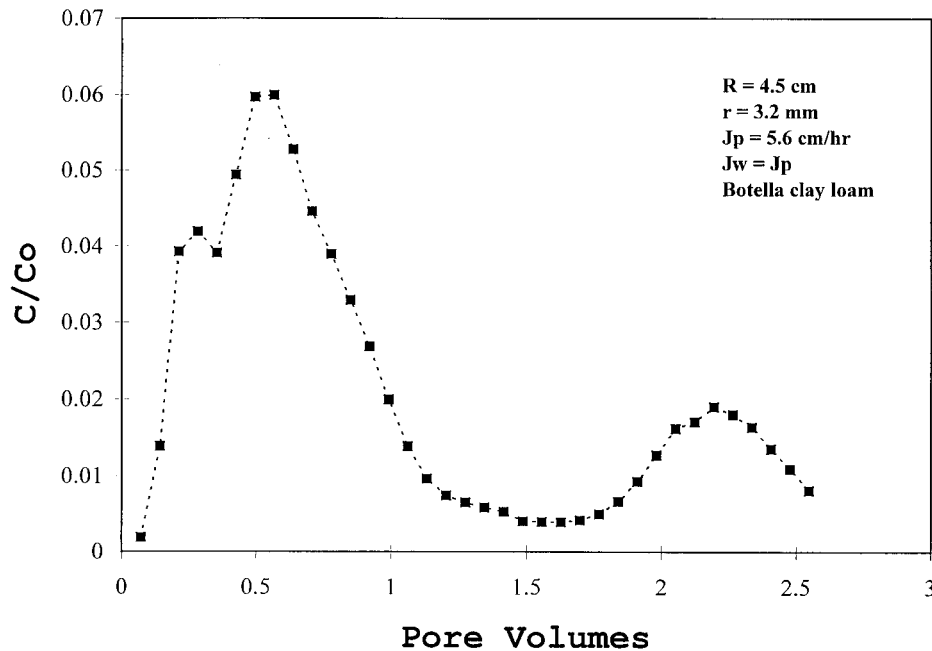


Fig. 3. Effluent breakthrough curve from a large matrix (R), small macropore (r) column of a Botella clay loam at J_p with a 1-cm pulse of bromide.

Macropore Flow Through a Clay Loam Soil

Miscible displacement experiments performed with the Botella clay loam soil illustrate that the observed macropore induced flow processes are not unique to silica sand (Fig. 3). It is clear from this BTC that for a soil with a fine texture, a low hydraulic conductivity, and a complex pore network in its matrix (compared with silica sand), the relative amount of water preferentially transported through the macropore (under steady flux conditions and at J_p) represents a larger fraction than transported through the matrix. These findings are in agreement with those of Li and Ghodrati (1994, 1995, 1997) who demonstrated that macropore flow occurred in soils with different textures, and that the relative macropore flow was always largest for the finest textured medium. The mean residence time for the second peak is greater than what would be expected had there been no macropore flow based upon the input flux. Since the total flux is a composite of the matrix and macropore flow, it is reasonable to see a later breakthrough time for the matrix zone which has a smaller relative flux.

The emphasis of this study is to examine the mechanistic processes governing macropore flow, rather than studying transport through the matrix itself. Taking these considerations into account, we performed the remainder of our experiments with the faster hydraulically conducting silica sand.

Visual Observations of Macropore Flow

The colored dye miscible displacement experiments illustrate the dependence of macropore flow on flux rate (Fig. 4). At the flux rate of J_p (24.1 cm/h, Fig. 4a), flow at the macropore/matrix interface was extensive and began as soon as the dye front reached the top of the macropore. By 0.06 pore volumes (V/V_0), the dye

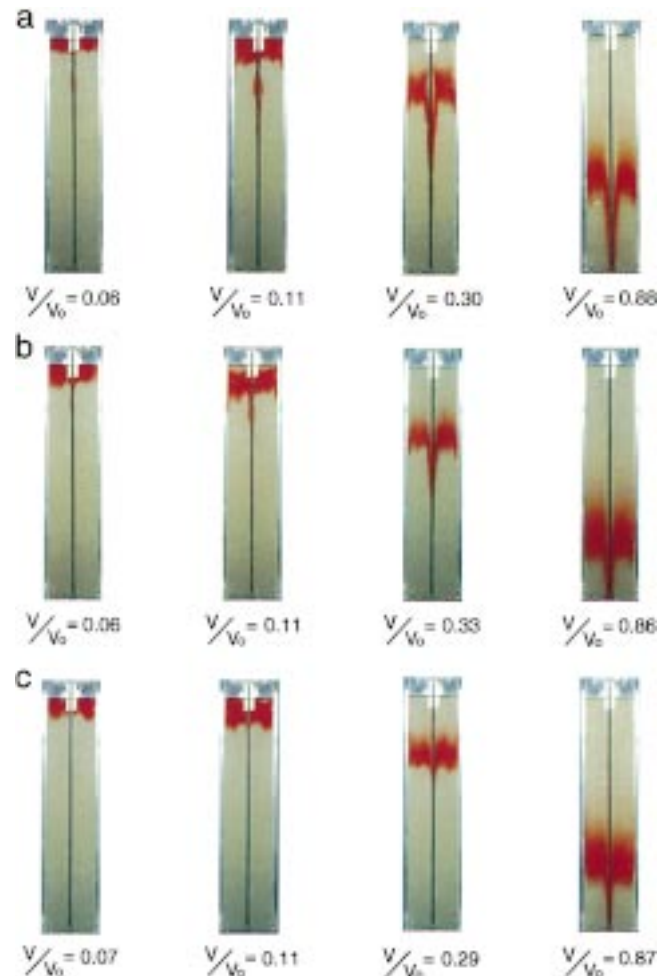


Fig. 4. Miscible displacement experiments in a large matrix (R), small macropore (r) column of silica sand with red dye at J_w = a) J_p , b) 66% J_p , and c) 33% J_p .

had traveled the entire length of the macropore and was observed exiting the column. As the dye front moved through the column, dye continued to appear in the macropore effluent. Furthermore, as the dye in the macropore moved downward from its point of entry, dye also entered the matrix directly adjacent to the macropore. This “meandering” of the macropore dye developed into a second zone where its transport was still accelerated relative to the rest of the matrix, but retarded relative to flow at the macropore. Finally, the remainder of the dye plume moved uniformly through the unaffected portion of the matrix.

As the flux rate was reduced to 66% of J_p (15.9 cm/h), the relative amount of macropore flow also appeared to decrease (Fig. 4b). This assessment was based on the observed decrease of dye transport in the matrix zone adjacent to the macropore. However, at both of these flux rates, the time and location of the macropore flow initiation remained unchanged. At a flux rate of 33% J_p (8.0 cm/h), direct transport through the macropore was no longer visible and the amount of dye transported through the matrix adjacent to the macropore was also markedly reduced. In fact no macropore flow was visible until 0.30 pore volumes had passed through the column. This initiation occurred prior to 0.06 pore volumes at the higher flux rates.

From our visual observations, it was clear that macropore flow occurred predominantly at the matrix–macropore interface and in the matrix zone directly adjacent to the macropore. Along this interface, only film flow was observed. Only at higher fluxes, where the system was ponded, was excess flow into the macropore (i.e., running water) observed. This was likely a result of the matrix–macropore interface film increasing in thickness to the point where excess flow became visually apparent. This supports the work of Wan et al. (1996) who showed an increase in fracture film thickness as the magnitude of the matric potential increased. As a result of these observations, we will define macropore flow in our system as flow through the macropore, flow at the macropore/matrix interface, and accelerated matrix flow in the matrix directly adjacent to the macropore.

In the zone of accelerated matrix flow, both diffusion and advective flow contribute to the entrance and movement of the tracer through this zone. As dye moved along the matrix–macropore interface, the presence of a concentration gradient likely drove the dye into the matrix zone. It is interesting that dye diffuses into the matrix only at certain positions along the interface. This may be a result of differences in entrapped air and matric potential along this interface. Also, dye does not appear to move into this transition zone from the surrounding matrix. This is supported by Fig. 4 since, at later times, dye is washed out of the matrix zone adjacent to the macropore. Wan et al. (1996) also observed this transitional zone between the matrix and macropore. They explained that the transport patterns in their glass micromodels depicted a decrease in advective transport and increase in brownian diffusion with increasing distance away from the fracture into the matrix zone.

The BTCs produced in a separately packed column with a nitrate tracer indicated that macropore flow disappeared near 70% J_p (18.3 cm/h) (Fig. 5). Other miscible displacement experiments using nitrate in the same column showed the disappearance of macropore flow to range from 70 to 55% J_p (data not shown). One explanation for this difference is that the exact amount of macropore flow will vary between replicates (on the basis of differences in entrapped air, particle arrangements, etc.). However, another possibility is that macropore flow occurred in the BTC experiments at flux rates less than 70% J_p but was undetected. Since the nitrate transported by the macropore is significantly diluted by the effluent, its concentration may have produced a signal less than the background light absorbance. Li and Ghodrati (1997) show similar breakthrough experiments where the macropore contribution

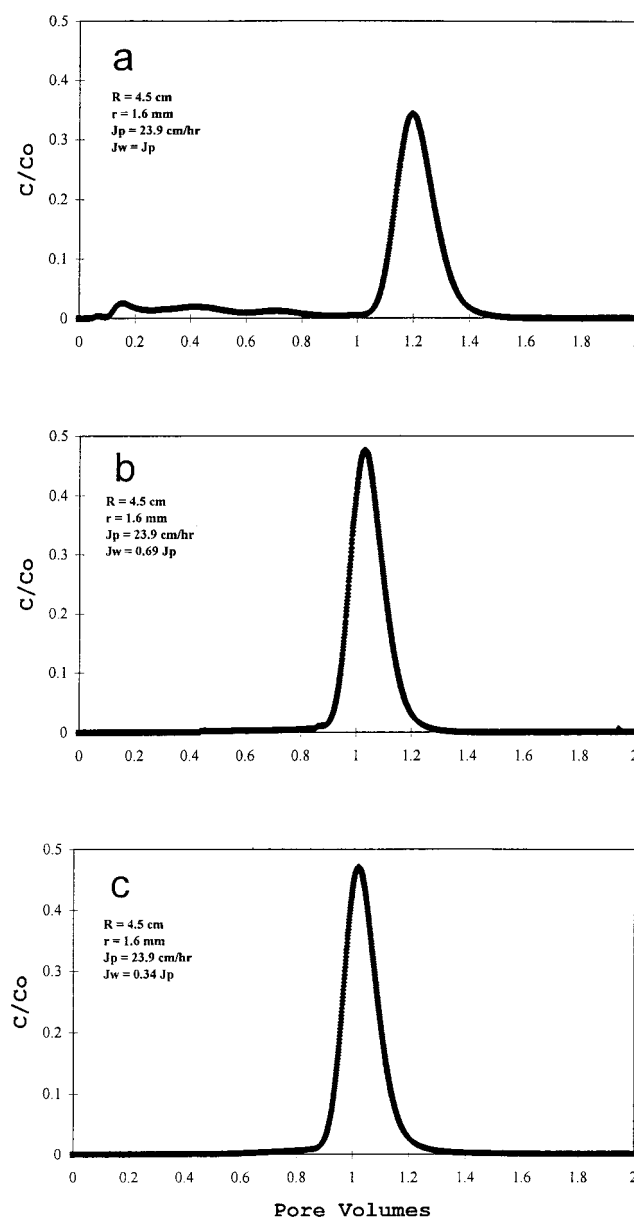


Fig. 5. Effluent breakthrough curves from a large matrix (R), small macropore (r) column of silica sand with a 1-cm pulse of nitrate at J_w = a) J_p , b) 66% J_p , and c) 33% J_p .

was a small fraction of the total effluent. This is also one possible explanation of why Czapar et al. (1992) did not measure any bimodal BTCs in their subsurface macropore experiments. If possible, separate collection of macropore effluent and matrix effluent should alleviate this problem.

Figure 6 depicts BTCs for all four SMCs under 1 cm of ponded head. Although the pattern of macropore flow through the SMCs was quite variable, all four columns have a three region BTC. The first region is likely a result of flow directly through the macropore/matrix interface. This explanation is supported by Fig. 4a, where at J_p , dye has diffused from the beginning of the macropore to the surrounding sand matrix at 0.06 pore volumes. At this time, dye was observed in the column effluent. The second zone may be attributed to enhanced conductivity near the macropore. Both Fig. 4a and b illustrate this assertion, where a long and narrow plume of dye was observed moving along both sides of the macropore. The third zone of flow is probably a result of transport through the matrix. Similar to transport in the Botella clay loam (Fig. 3), it is reasonable to see a later breakthrough time for the silica sand matrix which has a smaller flux relative to the average flux rate of the entire column.

The Impacts of Macropore Size and Matrix Size

Table 1 presents the variability in the determination of the J_p parameter. This variability is likely a result of differences in entrapped air and particle arrangement between different column replicates (Vandevivere and Baveye, 1992). The averaged J_p values in Table 1 show a trend in the effect of the macropore on the total flow regime. The small matrix columns have a greater J_p than the large matrix columns. However, this difference is significant only at an 80% confidence limit. On the basis

of the average J_p values, the effects of the two macropore sizes do not show any statistical trend.

A number of miscible displacement experiments were performed in the four columns at fluxes below J_p , at J_p , and at 1-cm head. We chose to present the 1-cm head experiments since the contribution of macropore flow to the breakthrough curve is enhanced. All the miscible displacement experiments we performed had similar breakthrough curve patterns. However, the relative contributions of each macropore associated breakthrough peak decreased as the amount of macropore flow decreased. Figure 6 suggests that, although there is certainly an intrinsic variability between columns, there were some interesting similarities. The transport patterns in the breakthrough curves at 1-cm head all exhibited at least three peaks. The earliest peak, which had an arrival time ranging from 0.03 to 0.14 pore volumes (average 0.06), corresponded to transport directly through the macropore zone. The final peaks were likely produced by transport through the matrix. The peaks between the initial and final peaks (Fig. 6a) were likely the accelerated matrix flow zones directly adjacent to the macropore. Figure 6 shows that these “middle” peaks had a small but definite contribution to the breakthrough curve. Moreover, mass recoveries for all miscible displacement experiments were complete with an average of 103% recovery and 6% coefficient of variability.

The breakthrough curve patterns in Fig. 6 also exhibit a great deal of variability. It is difficult to determine the source of variability in these flow patterns; however, differences in matric potential, surface roughness, and entrapped air near the matrix–macropore interface may be responsible. These factors have been shown to be influential to macropore flow (Phillips et al., 1989; Edwards et al., 1992; Shipitalo and Edwards, 1996; Wan

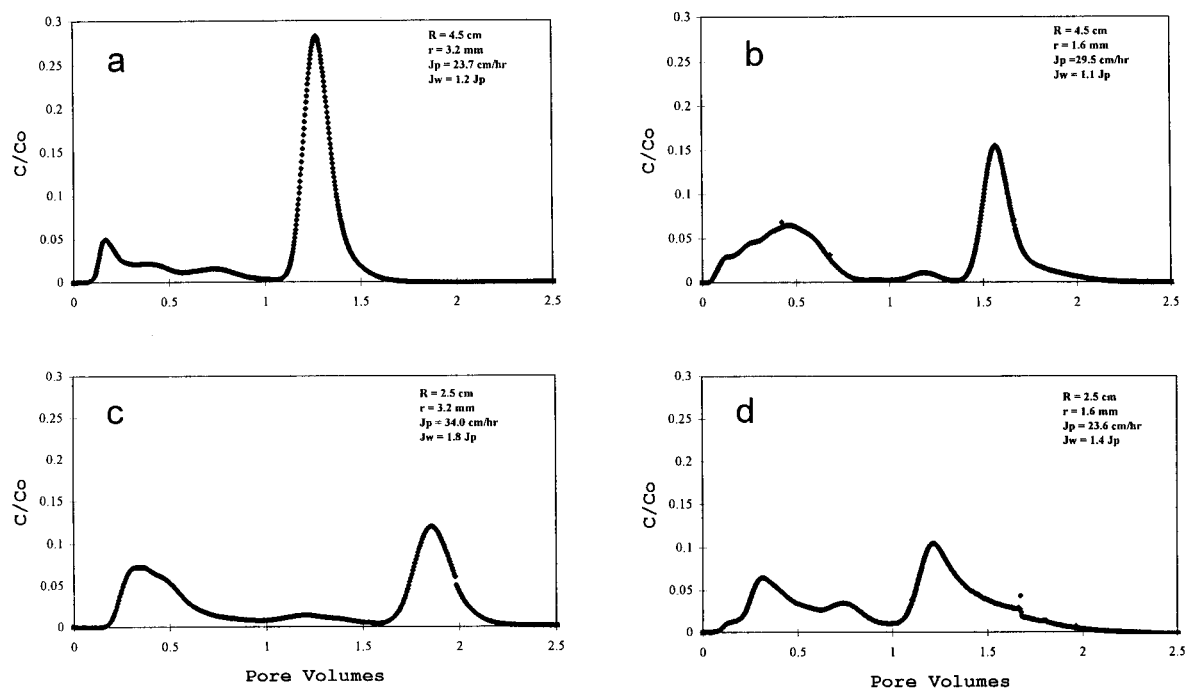


Fig. 6. Effluent breakthrough curves of silica sand at 1-cm head with a 1-cm pulse of nitrate in a) large matrix (R), large macropore (r), b) large matrix (R), small macropore (r), c) small matrix (R), large macropore (r), and d) small matrix (R), small macropore (r) columns.

et al., 1996; Tokunaga and Wan, 1997). However, we were unable to determine significantly their relative influences. Further studies aimed at quantifying these mechanisms would be valuable.

Through qualitative evaluation of these graphs, the relative amount of mass transported as a result of “macropore flow” (i.e., interface and transitional flow) was determined. The amount of mass transported as a result of macropore flow at 1-cm head was 38% of the total mass in the large matrix columns and 50% in the small matrix columns. Macropore transport was 40% in the large macropore columns and 48% in the small macropore columns. Along with the J_p data, these comparisons suggest that, in these column sizes, the ratio of the matrix to macropore size may be more influential in determining the quantity of macropore flow than the actual size of the macropore.

During these experiments, only a film of water was visible on the macropore surface. Subsequently, the macropore never became completely saturated. This suggests that for different macropore sizes under similar flux conditions, the absolute quantity of mass transported by the macropore should remain unchanged. Only the relative proportions of mass transport between the matrix and macropore should change. A more direct examination of the influence of macropore size is needed to validate this hypothesis.

SUMMARY

The split macropore column is a convenient tool which can be utilized to study the interactions between a macropore and a soil matrix. With this system, we observed the initiation and development of macropore flow under different flux rates and with different macropore to matrix size ratio configurations. From these observations, we noted three zones active during macropore flow. The first being flow directly at the matrix–macropore flow interface. The second was flow in the matrix directly adjacent to the macropore. This zone can be thought of as a transition zone between the gravity driven macropore/matrix interface and the convective/dispersive behavior of the third zone. The third type of observed flow occurred in the remaining portion of the matrix. This zone was largely unaffected by the presence of the macropore and produced uniform patterns of water flow through the column.

Quantification of macropore flow was made difficult because the macropore effluent was diluted by the matrix effluent. Therefore, future quantitative studies using this system should separate the matrix and macropore effluents. This modification would also allow researchers to separate information regarding transport through the macropore zone and transport through the matrix. Then data could be compared by means of variables such as mass balance, transport time, and breakthrough variance. This experimental system will permit examination of many issues raised by our initial data.

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DIVISION S-1—NOTES

The Effect of Hysteresis on Microbial Activity in Computer Simulation Models

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Abstract

Microbial activity in soils depends on the status of the soil water, which is expressed by pressure head (h) or water content (θ). There is no unique relationship between θ and h because moisture relations exhibit hysteresis. For convenience microbial activity has usually been related to the main drying curve but in general this will lead to bias in computer models that aim to simulate microbial activity. This article aims to evaluate the magnitude of the bias resulting from the use of computer simulation models of mineralization coupled to a model that follows the hysteresis of water relations in soil. Simulations of mineralization were found to accumulate a bias of more than 15% of the mean annual mineralization by harvest. This bias was found to be most serious where soil is continually subject to severe wetting and drying cycles such as can be found in irrigated and tropical agriculture.

THE RELATIONSHIP between the volumetric water content (θ , L^3L^{-3}) of a soil and the pressure head at which it is held (h , L) differs depending upon whether the soil is wetting or drying; this path dependence is known as hysteresis. The $\theta(h)$ relationship is characterized by a main drying curve and a main wetting curve but the actual state of a porous medium may lie within or on these envelopes. During wetting and drying cycles there are an infinitely large number of pathways that can be followed, known as scanning curves (see e.g., Topp, 1969). Fortunately, empirical and mechanistic

models exist that can describe scanning curves. In most experimental studies the main drying curve only is determined, but this curve always overestimates θ for any given value of h . For convenience, microbial activity is measured only in relation to this same drying curve; Stanford and Epstein (1974) and Stott et al. (1986) showed how nitrogen mineralization changes in relation to h . Groffman and Tiedje (1988) also observed the effect of hysteresis in moisture relations on denitrification. Computer modelers have made extensive use of the relationships between mineralization and either θ or h to simulate the change in microbial activity in soil with drying. Like the experiments, the models fail to take into account the effect of hysteresis on the moisture relations in soil.

This note aims to evaluate the magnitude of the bias resulting from the use of computer simulation models that calculate mineralization without including the effects of hysteresis. Clearly there are no soils that do not exhibit hysteresis yet their existence is automatically assumed by models that relate microbial activity to laboratory measurements made on the main drying curve only. Predictions of mineralization may be in error from day to day depending on whether the soil is wetting or drying. Here we demonstrate how these errors may be estimated and assess their likely magnitude.

Methods

Climate and Soils Data Used in the Models

Real, daily meteorological data close to the 40-yr mean for Wageningen in the Netherlands, longitude 5.67°E, latitude 51.97°N, was used for a series of simulations under temperate conditions (see Whitmore and Schröder, 1996, for a description of how the weather data was derived) and weather from Sanpatong, Thailand, longitude 98.95°E and latitude 18.75°N for a series under tropical conditions. Three rainfall regimes were tried under temperate conditions: (i) natural (unaltered), (ii) allowing 75-mm rain to fall each month in the first 5 d and none for the remainder of that month, (iii) spreading each month's rainfall uniformly over every day of that month. Daily

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